

Anti-surge Valves for Dynamic Compressors

Follow this guidance to improve the selection and sizing of anti-surge valves for axial and centrifugal compressors

Amin Almasi
WorleyParsons Services Pty. Ltd.

One of the most significant limitations on the operation of dynamic compressors (axial and centrifugal compressors) is the low-flow limit known as “surge”. A sudden valve closing, such as for emergency shutdowns or for a suction/discharge valve, presents significant challenges to anti-surge valve sizing and selection. Anti-surge valve requirements depend to a large degree on the compressor type, the compression system arrangement, and particularly the volume of the discharge piping downstream of the compressor. Important aspects of anti-surge valve sizing and selection for plants in the chemical process industries (CPI), as well as examples, are discussed in this article.

The basics

At low flows, the performance of a dynamic compressor — particularly centrifugal and axial compressors — is limited by the occurrence of aerodynamic flow instabilities that could lead to a catastrophic failure of the compressor due to mechanical or thermal loads. The surge is an unstable flow situation that occurs when the flow is too low. When a dynamic compressor reaches its surge limit, the flow pattern through the compressor

collapses, and a sudden backward flow of gas occurs from the discharge to the suction side of the compressor.

This surge phenomenon can cause a reversal of the thrust loads (large dynamic/transient forces on the rotor), which can result in damages to bearings (particularly thrust bearings), seals and other parts of the compressor. For these reasons a surge — and particularly high-energy surges — should be avoided. Recent innovations in control technology, dynamic simulations, rotating machinery knowledge and control-valve design (the anti-surge valve) have made it possible to supply anti-surge systems that are capable of coping with rapid flow fluctuations and process gas variations.

A dynamic compressor, also known as a “turbo-compressor”, is generally provided with an anti-surge valve (Figure 1) in order to keep the compressor in a stable operating range. This is achieved by assuring a suction flow that is higher than the corresponding flow at the surge. An anti-surge valve is placed in the anti-surge circuit (a recycle circuit).

The main purpose of this valve is to avoid a surge and surge-related damages, particularly irreparable damages (Figure 2). The anti-surge system and the controller, which comprise



FIGURE 1. This picture shows an example of an anti-surge valve

both hardware and software, should be based on a design algorithm that operates very quickly when opening the anti-surge (recycle) valve to avoid the surge. However, this anti-surge system should be designed and programmed so as to not make the circuit (or the compression unit) unstable.

Centrifugal and axial compressors experience a surge whenever they are suddenly tripped. Controlling the head and total energy of the surge event, particularly a surge at the trip, is the main objective of dynamic simulations and anti-surge valve sizing and selection. Meaningful gains can be made by a better understanding of the interactions between the compressor, the anti-surge valve(s) and the facility piping and layout (including coolers, scrubbers, check valves, vent valves and more). Key parameters for anti-surge valve selection and sizing include the following: fast response, optimum size, high integrity, accuracy and noise abatement.

The anti-surge valve

The size of an anti-surge valve influences the amount of fluid that can be moved from the discharge side of the compressor to the suction side. Larger valves allow more flow. The speed with which an anti-surge valve opens

is also important in rapidly reducing the discharge pressure (the head and stored energy at the compressor discharge). One important trade-off in relation to anti-surge valves is that larger valves exhibit slower opening rates. There are certainly cases when a faster, smaller valve is better than a larger, slower valve, but in all cases, the anti-surge valve should be large enough to handle the flow supplied by the compressor at a sufficiently low differential pressure (head).

Another important factor is controllability. Larger valves are more difficult to control. The faster the valve can be opened, the more flow can pass through it and the more effective it is in surge prevention. There are limits to the valve opening speed, dictated by the need to control intermediate positions of the valve. There are some practical limits to the power of the actuator. The situation may be improved by using a valve that is boosted to open, combining a high opening speed for the surge avoidance with the capability to avoid oscillations.

During a surge event, the flow increases rapidly at first and then decreases more slowly as the pressure difference across the compressor — and consequently the anti-surge valve, too — decreases. With a relatively large anti-surge valve, the energy level at which the surge occurs is much lower compared to a smaller anti-surge valve. An anti-surge valve with a relatively large capacity (compared to the compressor's volumetric flowrate), but with the same opening time (compared to relatively small anti-surge valves), will allow the shutdown surge to be avoided (or the shutdown surge will occur at a lower head condition). On the other hand, a relatively small anti-surge valve could cause a more energetic and more potentially damaging shutdown.

The ability to quantify these effects and select a proper valve size by dynamic modeling can be very useful. Because of this, accurate simulations are encouraged in order to select an anti-surge valve with an optimum size.

The effect of valve flow coefficient. In a case study for a centrifugal compressor in a CPI plant, two anti-surge

valves with the same opening time (around one second) were evaluated and compared. The small anti-surge valve resulted in a drop in flowrate without much change in the pressure head. The shutdown surge occurred at a moderately high discharge pressure. The large anti-surge valve — with a valve flow coefficient (C_v) about two times that of the small valve — resulted in a major reduction in the head. The shutdown surge occurred at a discharge pressure just above the suction pressure. The surge event in this case was not even noticeable. The flow reversed through the compressor at a low head point. This case study shows the surge-preventing effects of an anti-surge valve with a higher C_v compared to a smaller valve with a lower C_v .

Additional requirements for anti-surge valves include the following:

1. The reduction of stroke time (in response to control signal step changes)
2. A stable response
3. Minimized overshooting during the valve adjustment steps

The requirements for an anti-surge valve should always be optimized. Too fast a response can result in excessive overshoot and poor accuracy. However, too slow a response could result in sluggish opening of the anti-surge valve and possible inadequate protection of a compressor from a surge. Most information required for the sizing of anti-surge valves is available on the compressor map (the compressor curves). Well-known compressor manufacturers and anti-surge system suppliers have their own standards for the safety margins on anti-surge valve capacity. This figure is usually based on the compression system dynamics, the valve response and the system characteristics.

The anti-surge valve should be capable of passing 100% of the surge flowrate at around 50% of the valve opening. In other words, the anti-surge valve's C_v is selected to be approximately two times that of the required C_v , based on the compressor surge flow on the highest compressor speed curve (on the compressor map).

Valve noise. When the anti-surge valve is opened by the anti-surge sys-

tem, the valve dissipates a large portion of operating power of the compressor. A portion of the power dissipated by the valve is converted into acoustic energy, which becomes valve noise. Apart from the noise, considerations should also be given to the fluid velocity in the valve outlet. This should be kept within some limits to avoid pipe vibrations. Typically, CPI plants use an anti-surge-valve noise limit range (before the external attenuation) from 85 to 100 dB(A). The anticipated noise level (before the external attenuation) should not exceed 105 dB(A) with fluid velocities below Mach 0.3.

Anti-surge valves should be designed to increase the total system efficiency over a wide range of the operation. The inline and symmetrical flow path eliminates indirect flows and unnecessary changes in the flow directions through the valve. Axial-flow anti-surge valves are most often specified. This design is rugged and offers high performance. It is highly reliable and requires little maintenance. The "breaker vanes" are often used in the downstream section of the valve body (downstream of the valve's flow-control internals), which cut and streamline any flow turbulence. This can result in significant reductions in the noise and the turbulence, eliminating excessive vibrations in the valve and the associated piping, supports and structures.

The anti-surge valve should be capable of changing its position in a short period of time. For reasons of reliability, a spring-loaded actuator is desirable. The typical control instrumentation and accessories provided on an anti-surge valve can include a valve positioner, a volume booster and a solenoid. In other words, usually a solenoid valve with a well-designed booster device (that accelerates the positioner operation) is used.

The correct size and configuration of the required actuators, instrumentation and accessories could guarantee an anti-surge-valve response time of less than 2 s (to fully open). An overshoot on intermediate changes should be kept to an absolute minimum.

Control range. The anti-surge valve should have a wide control range. The whole valve system should operate



FIGURE 2. Axial compressors are very vulnerable to damage from surges. This photo depicts the type of blade damage that can be expected as a result of a surge event in an axial compressor

very quickly in case of an emergency (particularly an emergency-trip surge event). A very high operating rangeability is usually required. Typically, a rangeability of 150:1 is specified. The rangeability parameter is defined as the ratio between the rated C_v for a completely open valve and the minimum C_v that the valve can control. A high rangeability value means successful control even with a high differential pressure and a low flow.

To increase the rangeability, special trims are used where the area distribution is made according to the valve characteristic curve. For example, in some valve designs, until 40% of the stroke, the C_v is typically limited to 15–25% of its maximum value to obtain superior controllability. Too often, a special trim with multiple jumps is specified for an anti-surge valve.

Service reliability. Good service reliability even under very severe conditions (for example, compression ratios lower than 0.3, a discharge pressure ranging up to 600 bars or more, and other difficult situations) is required. In order to obtain these requirements, the following considerations should be respected:

- Anti-surge valves in various trim styles are usually fitted with pressure balanced pistons. The required thrust should be virtually independent of the differential pressure across the valve
- Bushings should be anti-seize and self-lubricated. The internal sliding area requires high-quality and special design and materials (usually chromium-plated internals are specified)
- Proper packings should be used. Typically, a charged polytetrafluoro-

ethylene (PTFE) with a special design is specified. For low molecular-weight gases (such as hydrogen) high-quality graphite packing is commonly specified

- The trim materials should be chosen for good corrosion-proofing and erosion-proofing. For usual services, a proper stainless-steel alloy is typically specified. For a special application, a special, sintered tungsten carbide could be a good selection. The trim design should allow for excellent sealing
- For high-pressure anti-surge valves, additional requirements should be considered. For example, the welded seat is replaced by a high-quality cage seat for superior sealing, better centering and easier maintenance and operation

Leakage. Leaks across an anti-surge valve can influence the efficiency of a compressor system. Even under the most severe working conditions, high-quality anti-surge valves should maintain a tight shut-off (within practical limits) over the full pressure range. With this feature, the compressor efficiency is maintained at the highest levels during the normal operating mode. In a case study for a medium-size high-pressure centrifugal compressor (8-in. ASME Class 900 recycle control valve), with an ASME Class IV leakage rate and a differential pressure of around 70 bars, the valve leaked approximately 150 Nm³/h of a compressed gas.

The leakage feature should be independent of the actuation method. New designs do not need a higher torque as was required in older designs. In modern designs, the ASME Class IV sealing class is usually achieved by the

position and not by the torque. Rapid changes in the differential pressure across the anti-surge valve should have no effect on the stability of the valve position. A fast response is usually achieved with properly sized actuators in contrast to those fitted on conventional control valves. These advanced actuators are usually very efficient and compact. They cannot be bulky or massive. Modern designs contain a minimum number of moving parts and a short valve stroke. These advanced designs permit fast actions and reliable operation.

Various studies, different simulations and extensive operational experiences pertain to a typical “one anti-surge valve, one compressor casing” arrangement. More complex systems of cascaded valves (or valves around multiple compressors) require a more detailed analysis and sophisticated provisions.

Valve and size selection. An anti-surge valve should be sized to meet two diverse objectives: steady-state operation and transient cases.

During steady-state recycling, the required capacity of the anti-surge valve can be directly derived from the compressor map (the compressor curve). Typically, smaller valves exhibit smoother control. During transient conditions, however, the required valve capacity increases. To avoid a surge during an emergency shutdown (or similar transient events, such as a sudden suction or discharge valve closing, or others), a bigger valve will give better performance. To fulfill both above-mentioned requirements, an anti-surge valve with an equal percentage characteristic (defined below) is recommended. Two types of anti-surge valves are generally used: globe valves and ball valves (noise-attenuating ball valves).

The equal percentage characteristic spreads the first half (50%) of the valve’s fully open capacity over the first two thirds (66%) of the valve’s travel for a globe valve, and about one third (33%) of the valve’s fully open capacity over the first two thirds of the valve’s travel for a ball valve. An anti-surge valve should be described by its maximum capacity (C_v), and by its capacity as a function of the valve

travel, and the opening behavior. The globe valve's C_v approximately varies with the square of the percentage travel. The ball valve's capacity varies roughly with the cube of the percentage travel. A ball valve will have more capacity to depressurize the discharge volume compared to a globe valve of the same size.

In a case study for a 6-in. (150 mm) anti-surge valve, the C_v of a ball valve was more than 2.5 times of the C_v for a same-sized (6-in.) globe valve. At two thirds of valve travel, the ball valve flowrate was more than 50% higher than that for the same-sized globe valve. This additional flow capacity would sometimes, theoretically, make the ball valve a better choice in an anti-surge application. This is particularly true where there is only a single anti-surge valve per stage (for example, there is no hot-gas bypass) and the discharge volumes are large (especially when the anti-surge loop is taken from downstream of the discharge cooler). However, this is just a theoretical and textbook idea.

From a practical standpoint, the globe valve behavior is more predictable. The globe valve is more comfortable to control and manage. It is relatively less nonlinear. The ball valve usually offers a highly nonlinear behavior (C_v varies with the cube of the percentage of travel).

The anti-surge valve should be matched to the compressor system. Both globe and ball valves are used in modern anti-surge systems (depending on the application). Practically, a globe valve is more common in anti-surge systems.

In some compressor installations, it is necessary to have both a normal anti-surge valve (the recycle line taken from downstream of the after-cooler or the anti-surge line that includes a cooler) and a hot-gas bypass valve, which can be opened to rapidly transfer the compressed gas back to the suction side of the compressor without cooling (usually for a short period of time). A hot-gas-bypass allows the pressure head across a compressor to be rapidly reduced during a trip. Various options for anti-surge valve size, opening rate, recycle loop arrangement and operation sequence

should be evaluated. Parametric studies of the anti-surge valve size and opening time can yield optimum sizes for an anti-surge valve (and a hot-gas-bypass valve, if required).

Overheating. A large portion of the mechanical energy produced by a compressor is converted into heat in the discharged gas. In an uncooled recycle-loop system, this heat is recycled into the compressor suction end, and then more heat energy is added to the recycling gas. In a typical natural-gas centrifugal compressor at a 100% recycle without recycle cooling, the temperature of the gas increases about 1°C per second. Eventually, this could lead to overheating at the compressor discharge side.

In theory, the recycling can lead to heating whenever the compressed gas cannot be cooled to the initial operating temperature at the compressor suction side. This is the case both for a hot-gas bypass and a partially cooled bypass (if the cooled bypass is not designed properly, for example, when sufficient cooling capacity is not provided). The overheating could be defined as the gas temperature within the compressor exceeding 150–190°C, depending on the process. The same problem may also occur if there is a long period of time between the start of rotation (the compressor startup) and overcoming the pressure downstream of the discharge check valve, when some kind of hot-gas bypass is used for the startup.

Control and testing

An anti-surge-system configuration usually allows an operator to open the anti-surge valve further than the valve position that is defined by the anti-surge controller (based on the algorithm) in manual-mode operation. However, the operator cannot close the anti-surge valve any further than the surge controller permits. The controller can override the manual mode, if necessary, to open the anti-surge valve. The “fail safe” philosophy that is commonly employed is that the anti-surge valve should be an “air-to-close” valve (fails open). The control is used to force the anti-surge valve fully open whenever the compressor is shutdown. This helps to prevent a

surge on a compressor shutdown, and also holds the anti-surge valve in a fully open position for a startup.

Extreme care has to be taken when testing an anti-surge system during operation. The compressor should not accidentally be pushed into a surge. The controller should take over if something causes the operating point to head toward a surge. Such features should be available, enabled and working correctly. A sudden movement of the anti-surge valve could have an adverse impact on the remaining gas system (the upstream and the downstream of the compressor). All of these control actions should be evaluated properly.

It could be useful to assess how the anti-surge valve behaves in response to a small 1–2% change in the command signal. Bad signs are hunting, stiction (the static friction that needs to be overcome when the anti-surge valve-position should be changed), hysteresis and delayed lifting, all of which could make the control more difficult.

The anti-surge step test should be performed with the controller in the manual mode and with the valve off its seat, preferably somewhere between 15–85% open. The compressor will be on partial-recycle throughout the test (the remaining process should be able to cope). The first valve step could probably be in the open direction so as to move the compressor further from the surge. If the response is unknown, a small valve movement (even typically 1–2%) should be selected.

One important consideration when operating in the recycle mode is that the gas could thin out due to loss of the heavier components. The surge line of the lower molecular-weight gas may be rather different from the normal gas. Some amounts of new gas should always be fed through the compressor. Another useful test is to introduce a disturbance to the compressor, such as cutting back the suction throttle valve slightly or changing the driver speed (if it is a variable-speed compressor train). It should be verified that the anti-surge valve changes position without any delay and also without a significant undershoot of the setpoint.

Examples

Example 1. The first example is presented for a two-process-stage centrifugal compressor. The two process stages are low pressure (LP) and high pressure (HP). The compressor is a variable-speed synchronous electric-motor-driven unit (using gears) to compress natural gas from around 10 barg to approximately 100 barg. Each compressor stage has an anti-surge valve.

The calculated C_v for the LP and HP compressor casings (at the surge flow of the high-speed curve) are 215 and 81, respectively. The selected anti-surge valve's C_v values for the LP and HP stages are 385 and 169, respectively. In other words, the C_v factors (the anti-surge valve C_v / the calculated C_v at the high-speed surge flow) are approximately 1.79 and 2.09 for the LP and HP loops, respectively.

The selected anti-surge valves are globe valves. The opening time is 2 s and the closing time is 5 s. The valves were purchased with a NACE (National Association of Corrosion Engineers)-compliant carbon-steel body, a stainless-steel (grade 17-4-PH) trim and special PTFE packing. Grade 17-4-PH stainless steel is a precipitation hardened (a chromium-copper precipitation hardened) stainless steel, employed for applications requiring high-strength and a high level of corrosion resistance.

Example 2. The second example is presented for a petroleum-refinery type hydrogen-service centrifugal compressor. It is a steam-turbine-driven, hydrogen-recycle centrifugal compressor for a hydrocracker unit to compress the hydrogen from around 140 to 190 barg. The calculated C_v for the surge point at the high-speed curve (on the compressor map) is 179. The recommended range for the anti-surge C_v is 1.7–2.1 times the calculated C_v for the maximum-speed surge flow. Applying the 1.7 factor (the minimum factor) results in an anti-surge valve C_v above 305.

The selected anti-surge valve is an 8-in. globe valve, 1500# class, RTJ (ring type joint) facing, with a C_v of 308, an opening time of 2 s and a closing time of 6 s. The anti-surge valve was selected with a carbon-steel body (a NACE grade), a stainless-steel (grade

17-4-PH) trim and a special graphite packing. Since the compressor train inertia is low (it is a steam-turbine, direct-driven compressor train), the anti-surge loop is taken from the immediate downstream of the compressor (upstream of the cooler).

Based on experience, for a compressor configuration like this, a relatively low C_v (around or lower than 1.7) is sometimes specified by vendors. Some vendors used C_v factors (the anti-surge valve C_v divided by the calculated C_v at the high-speed surge flow) even as low as 1.55 or 1.6 to keep the anti-surge valve size small. This is not recommended, however, since a sufficiently large anti-surge valve is necessary to handle emergency rapid shutdown situations and unscheduled trips at uncommon (off-design) operating cases.

In other words, 1.7 is the minimum acceptable factor. This hydrogen compressor train is expected to compress nitrogen during the initial startup

of the unit. The calculated C_v for the nitrogen operation (the surge flow at the high-speed curve) is around 260. In other words, a surge could be expected in case of an unscheduled rapid trip during the operation with the nitrogen, if a small anti-surge valve is selected. ■

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Author



Amin Almasi is a lead rotating equipment engineer at WorleyParsons Services Pty Ltd. in Brisbane, Australia (amin.almasi@worleyparsons.com). He previously worked in Técnicas Reunidas (Madrid) and Fluor (various offices). He holds a chartered professional engineer's license from Engineers Australia (MIEAust CPEng-Mechanical), and a chartered engineer certificate from IMechE (CEng MIMechE), RPEQ (Registered Professional Engineer in Queensland). He also holds M.S. and B.S. degrees in mechanical engineering. He specializes in rotating machines including centrifugal, screw and reciprocating compressors, gas and steam turbines, pumps, condition monitoring and reliability. He has authored more than 45 papers and articles dealing with rotating machines.

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